

WISRIGHT

PROJECT REPORT

AI-Based Inventory Movement & Stock Monitoring System

RFID + AI Camera Verification for Auditable, Real-Time Inventory Movement

Prepared for Internal Review

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1. Overview

1.1 Summary

An RFID and AI camera verification system that automatically counts and verifies products as they move out of warehouse racks, eliminating manual stock-counting errors and providing a real-time, auditable view of inventory movement.

1.2 Problem Statement

Manual stock movement today relies on paper logs or basic RFID scans that record what a worker claims to have taken, with no independent check on actual quantity. This creates the following issues:

- Human error in counting
- Mismatched inventory counts
- Delayed detection of stock discrepancies
- No independent verification of manual entries

1.3 Proposed Solution

Add an automated AI verification step so every manual RFID scan is cross-checked against an actual camera-based count, via an RTDETR AI model, before it is accepted into the central database.

1.4 Goals

- Eliminate manual stock-counting errors through AI-verified counting.
- Create a live, time-stamped audit trail of every stock movement.
- Give management a real-time dashboard view of inventory health.
- Improve worker accountability without adding significant extra work.

1.5 Non-Goals

- Replacing RFID/manual scanning entirely — the system verifies manual entries rather than eliminating them.
- Full computer-vision identification of product type on open racks — verification is confined to the enclosed tray to control lighting and clutter.
- Automated procurement or reordering workflows — out of scope for this phase.

2. Target Users & Applications

2.1 User Roles

Role	Description	Primary Interface
Warehouse Worker (Labour)	Picks products from racks, scans RFID tags, places items in verification tray	Mobile / Tablet Guide
Supervisor / Management	Monitors stock levels, movement trends, and mismatch alerts	Web Dashboard
System Administrator	Manages hardware, integrations, and user/rack configuration	Web Dashboard (admin)

2.2 Target Industries

Warehouses, Factories, Retail Stores, Pharmaceutical Storage, E-commerce Fulfilment, and Logistics Hubs.

3. End-to-End Workflow

The system implements a five-step movement and verification flow:

1. Labour Entry — Worker enters the stock room; an RFID reader scans their ID tag and logs entry time automatically.
2. Rack Identification — Each rack has its own RFID scanner and a QR code (A, B, C, D...) showing which products are stored there.
3. Product Taken (Manual) — Worker picks the required product and scans its RFID tag; entry is logged with labour name, rack, product, quantity, and time.
4. AI + Tray Verification — The picked product is placed inside a transparent verification tray. An RTDETR AI camera counts the items and compares the AI count against the manual count.
5. Data Stored — Once the AI count matches the manual count, the verified record (labour, rack, product, quantity, time, status) is saved to the central database. A mismatch raises an alert for manual review instead of auto-storing.

4. Functional Requirements

4.1 Labour Entry Tracking

1. FR-1: System shall scan an RFID tag assigned to each worker at the stock-room entry door.
2. FR-2: System shall automatically log entry time and worker identity on scan.

4.2 Rack Identification

3. FR-3: Each rack shall be fitted with an RFID scanner and a unique QR code (A, B, C, D...).
4. FR-4: Scanning a rack's QR code shall display the list of products stored on that rack to the worker via the mobile guide.

4.3 Manual Product Scan

5. FR-5: Worker shall scan the product's RFID tag when picking it from a rack.
6. FR-6: System shall log labour name, rack ID, product ID, quantity, and timestamp for every manual scan.

4.4 AI + Tray Verification

7. FR-7: The picked product(s) shall be placed into a transparent, enclosed verification tray.
8. FR-8: A top-mounted RTDETR AI camera, aided by built-in LED lighting, shall capture an overhead image of the tray.
9. FR-9: The RTDETR model shall detect and count each individual product in the image.
10. FR-10: System shall automatically compare the AI-detected count against the manually scanned quantity.
11. FR-11: On MATCH, the transaction shall be marked "Verified" and saved to the database.
12. FR-12: On MISMATCH, the system shall raise an alert and withhold the record from being auto-stored pending manual review.

4.5 Data Storage & Records

13. FR-13: Every verified record shall store: labour, rack, product, quantity, time, and status.
14. FR-14: All records shall be persisted to a central cloud/server database.
15. FR-15: The system shall maintain a centralised, queryable log/report history of all transactions (matched and mismatched).

4.6 Employee Mobile Guide

16. FR-16: Mobile/tablet app shall display each rack (A, B, C, D...) and the exact products stored there.
17. FR-17: App shall support QR code scanning to look up rack contents.
18. FR-18: App shall provide scan history, notifications, and worker profile management.

4.7 Web Dashboard

19. FR-19: Dashboard shall display total products in stock.
20. FR-20: Dashboard shall display items running low (low-stock alerts).
21. FR-21: Dashboard shall display today's total stock movements.
22. FR-22: Dashboard shall display active mismatch alerts.

23. FR-23: Dashboard shall provide stock-overview and movement-trend charts.

4.8 Hardware / Connectivity

24. FR-24: An ESP32 controller shall collect data from all scanners/cameras at a rack/tray cluster and transmit it over Wi-Fi to the cloud.

25. FR-25: A shared power supply line shall run across rack scanners and the verification tray per cluster.

5. System Architecture

5.1 Data Flow

RFID Tag (Labour) → RFID Reader (Entry Door) → Rack RFID Scanner (Each Rack) → Product RFID Scanner → RTDETR Camera (Verification Tray) → ESP32 Controller → Cloud / Server Database → Web Dashboard

5.2 Components

Component	Function
RFID Tag (Labour)	Identifies the worker at the entry door
RFID Reader (Entry Door)	Logs entry time and worker identity
Rack RFID Scanner	Confirms which rack is being accessed
Product RFID Scanner	Captures manual product scan and quantity
RTDETR Camera (Verification Tray)	Performs AI count for verification
ESP32 Controller	Aggregates scanner/camera data, sends over Wi-Fi
Cloud / Server Database	Stores every verified transaction securely
Web Dashboard	Live stock levels, alerts, movement trends
Mobile Guide App	Rack lookup, QR scanning, scan history, notifications

5.3 Transparent Verification Tray

- Enclosed, transparent tray with top-mounted RTDETR camera and built-in LED lighting.
- Purpose: remove background clutter, inconsistent lighting, and overlapping stock that would reduce AI counting accuracy on an open rack — maximizing match-rate reliability.
- Verification sequence: Image Capture → AI Detection → Count & Verify → Result (Match/Mismatch).

6. Non-Functional Requirements

Category	Requirement
Accuracy	AI product count must reliably match manual count under controlled tray lighting; mismatch rate should be validated during pilot.
Latency	Verification result (image capture to match/mismatch) should return in near real-time to avoid blocking worker workflow.

Category	Requirement
Reliability	ESP32-to-cloud data transmission must handle intermittent Wi-Fi gracefully (local buffering/retry).
Scalability	Architecture must scale across multiple racks, rooms, and warehouse branches without redesign.
Security	Worker RFID identity, stock data, and transaction logs must be stored and transmitted securely.
Auditability	Every transaction (matched or mismatched) must be fully traceable with labour, rack, product, quantity, time, and status.
Maintainability	Shared power supply and simple hardware chain should minimize installation/maintenance overhead.

7. Key Features Summary

- Labour entry tracking via RFID
- Rack identification using QR codes
- Manual RFID product scan logging
- AI camera verification using RTDETR
- Enclosed transparent verification tray
- Automatic mismatch alerts
- Real-time stock monitoring
- Centralised data logs & reports
- Employee mobile guide
- Secure, accurate, low-manual-error system

8. Success Metrics

- Reduction in stock discrepancies versus the current manual-only process.
- Percentage of transactions auto-verified (MATCH) versus flagged (MISMATCH) during pilot.
- Time-to-detection for stock discrepancies (should approach real-time).
- Dashboard adoption rate among supervisors/management.
- Reduction in time-to-locate products by warehouse staff (via mobile guide).

9. Rollout Plan

6. Pilot — Deploy on a single rack cluster to validate AI counting accuracy under real warehouse conditions.
7. Integration — Connect the ESP32 data feed with the company's existing inventory database.
8. Rollout — Deploy the mobile guide to warehouse staff and the dashboard to supervisors.
9. Scale — Expand tray verification points across all racks once pilot results are confirmed.

10. Open Questions / Risks

- What is the acceptable mismatch/false-alert rate before the system is considered pilot-successful?
- How are mismatches resolved operationally — who reviews, and what is the escalation path?
- What happens to throughput if every item must pass through a single verification tray per rack cluster (potential bottleneck)?
- Integration approach/API contract for the existing inventory database is not yet defined.
- Multi-item counting accuracy when products are stacked/overlapping inside the tray needs validation.
- Offline/degraded-connectivity behavior for ESP32 controllers is not yet specified.

11. Out of Scope (This Phase)

- Automated reordering/procurement triggers.
 - Open-rack (non-tray) AI product recognition.
 - Cross-warehouse inventory transfer workflows.
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